

OKINOERABU AP, Japan

WMO: 479420

Lat: 27.432N

Lon: 128.706E

Elev: 97

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	52.3	54.1	36.4	31.6	55.5	38.0	33.8	56.3	31.3	59.9	28.9	59.5	17.5	330	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	89.4	80.7	88.5	80.5	87.6	80.2	82.6	86.3	81.9	85.9	81.3	85.5	11.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.7	165.4	84.3	80.8	160.8	84.2	80.1	156.8	84.1	46.5	86.2	45.8	86.1	45.2	85.7	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
30.6	26.6	23.7	DB	49.0	90.9	2.0	1.1	47.5	91.7	46.4	92.4	45.2	93.0	43.8	93.8
			WB	43.3	83.3	1.4	1.2	42.3	84.2	41.4	84.9	40.6	85.6	39.6	86.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.9	61.6	62.1	64.6	69.0	73.8	79.0	83.8	83.9	82.1	77.7	71.7	65.0
	DBStd	8.78	4.30	4.56	4.49	3.74	3.09	3.40	1.81	1.89	2.32	3.15	3.73	4.35
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	356	124	103	64	9	0	0	0	0	0	0	2	54
	CDD50	8365	359	339	453	571	737	870	1048	1052	963	857	650	466
	CDD65	3246	18	22	52	130	272	420	583	587	513	392	202	55
	CDH74	27507	2	3	22	136	889	3418	6970	7151	5617	2771	503	25
	CDH80	8221	0	0	0	3	51	696	2614	2795	1685	365	12	0
(14) Wind	WSAvg	12.6	13.9	13.6	13.5	12.2	10.8	11.0	11.2	11.8	12.2	13.7	13.4	13.9
(15) Precipitation	PrecAvg	74.4	4.1	4.2	5.7	6.3	7.9	11.0	5.6	6.9	7.1	6.4	4.4	3.7
	PrecMax	127.9	9.9	15.6	14.2	17.1	18.0	28.1	18.1	25.7	15.5	21.8	8.5	9.3
	PrecMin	45.6	0.4	1.5	1.7	1.1	0.9	2.8	0.2	1.3	0.8	0.6	1.0	1.2
	PrecStd	18.5	2.3	3.1	2.7	3.3	3.9	5.4	4.9	5.0	4.4	5.4	2.2	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.6	74.1	76.7	80.0	83.8	88.7	90.4	90.8	89.6	86.4	81.7	76.6
		MCWB	67.1	68.6	71.3	75.2	78.7	81.6	81.1	81.2	80.9	78.6	75.0	70.0
	2%	DB	71.5	71.9	73.7	77.1	81.4	87.0	89.2	89.5	88.1	84.9	79.6	74.4
		MCWB	65.4	66.9	68.4	72.5	76.8	81.0	80.7	80.6	79.9	77.6	73.1	68.1
	5%	DB	69.8	70.5	72.1	75.4	79.5	85.5	88.4	88.6	86.9	83.3	78.1	72.8
		MCWB	64.1	65.6	67.3	70.8	75.0	80.5	80.5	80.4	79.2	76.4	71.8	66.4
	10%	DB	68.1	68.9	70.7	74.1	78.0	84.1	87.5	87.7	85.9	82.1	76.8	71.5
		MCWB	61.9	64.1	66.2	69.6	73.6	79.9	80.3	80.2	78.6	75.3	70.7	65.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.3	70.4	72.5	75.8	79.5	83.0	83.7	83.0	82.2	80.3	76.7	72.5
		MCDB	72.1	72.8	75.1	78.6	82.0	85.8	86.2	86.9	87.3	84.2	80.0	75.1
	2%	WB	67.0	68.5	70.6	74.0	77.8	81.9	82.7	82.3	81.3	79.1	75.2	69.7
		MCDB	70.3	71.0	72.6	76.2	80.2	85.2	85.9	86.6	86.0	83.2	78.3	73.0
	5%	WB	64.8	66.8	69.0	72.7	76.5	81.0	81.9	81.7	80.6	78.0	73.6	67.8
		MCDB	68.8	69.7	71.3	74.6	78.5	84.1	85.6	86.2	85.0	82.2	77.2	71.7
	10%	WB	62.9	64.8	67.3	71.1	75.3	80.2	81.3	81.0	79.9	76.9	72.0	66.0
		MCDB	67.6	68.5	70.1	73.2	77.2	83.2	85.4	85.7	84.3	81.1	76.1	70.8
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	7.1	7.5	7.3	7.1	7.0	7.3	7.3	7.0	6.5	6.3	6.5
		MCDBR	8.4	8.4	8.4	8.2	7.9	8.5	8.1	8.1	7.5	7.2	6.8	7.5
	5% WB	MCWBR	7.6	7.4	7.2	6.2	5.3	4.4	3.7	3.8	4.0	4.6	5.4	6.4
		MCWBR	8.0	8.1	8.0	8.0	7.4	8.1	7.4	7.4	7.0	6.7	6.7	7.3
(40) Clear-Sky Solar Irradiance	taub		0.456	0.487	0.551	0.558	0.516	0.476	0.470	0.483	0.480	0.475	0.460	0.450
		taud	2.118	2.055	1.908	1.948	2.137	2.328	2.345	2.300	2.275	2.225	2.204	2.140
	Ebn at Noon		244	247	239	242	252	260	262	258	256	249	242	239
		Edn at Noon	43	50	61	60	50	41	40	42	42	42	40	41
(44) All-Sky Solar Radiation	RadAvg		742	907	1186	1408	1551	1635	2106	1951	1674	1278	912	719
	RadStd		93	137	121	105	192	138	150	105	108	85	83	59

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	+0.56	+0.90	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	+0.67	+0.88	N/A	N/A	N/A	N/A

Nomenclature: See separate page